

Field-Circuit Coupling Using Black-Box Simulation Tools

Coupling to a Reduced-Order Field Representation via Waveform Relaxation

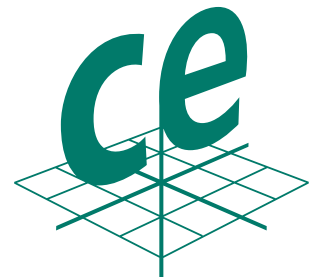
Bachelor thesis in the field of study "Computational Engineering" by Philip Simon

Date of submission: September 18, 2026

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Bei einer Thesis des Fachbereichs Architektur entspricht die eingereichte elektronische Fassung dem vorgestellten Modell und den vorgelegten Plänen.

Darmstadt, 18. September 2026

P. Simon

Abstract

Electric machines operate at relatively low frequencies yet are typically driven by high-frequency power-electronic circuitry. The step size required to solve either problem is determined by their individual time constants, which are inversely proportional to their frequencies. Since the time constants of these problems may be orders of magnitudes apart, they are most efficiently time-integrated at different rates. Their differing character – one a complex machine, modelled as a distributed device with differential equations and simulated numerically using methods such as the Finite Element Method (FEM), the other a circuit solved by its own solver – motivates a partitioned co-simulation rather than a monolithic approach. This work couples the domains with a Waveform Relaxation (WR) scheme, in which an inductance-based reduced-order model of the field serves as the interface condition and, as a linearisation term, accelerates convergence. The scheme is implemented with Xyce, Sandia's open-source SPICE simulator, leveraging its checkpoint/restart feature to re-solve each time window across WR iterations. The implementation is verified against monolithic solutions for cases with varying circuit complexity and a dummy field solver. A parameter study over [window length/linearisation update / ...] shows that [key result], from which [a concrete guideline, e.g. for choosing the window size] is derived.

Acknowledgements

I want to thank my supervisors, Prof. Dr. rer. nat. Sebastian Schöps (TU-Darmstadt) and Dr. Jörg Ostrowski (Siemens) for the support and guidance they offered me throughout this thesis. Sebastian Schöps' dissertation and papers on (field-circuit) co-simulation lay the mathematical foundation for this work and have been a constant source of knowledge. Jörg Ostrowski's idea of the *accumulated time step* was integral to this thesis. Thank you both.

Use of AI-based Tools

In Chapter 1 Claude (Anthropic) was used to improve the text quality and find suitable references. In Chapters 2 and 3 Claude was used to improve text structure and clarity, to reformulate passages I had drafted, and to point out inconsistent notation. In Chapter 3 it was additionally used as a sounding board during the derivation as it flagged candidate inconsistencies and questioned steps, which I then verified, corrected, or rejected myself. All mathematical results, derivations, and their correctness are my own and were checked by me. Grammarly was employed throughout to enhance writing quality, ensuring grammar, spelling, and style accuracy. Claude was used to improve grammar and phrasing in Chapter 4.

A rudimentary version of the C++ driver was originally written by myself and later extended massively using Claude Code, which I directed and reviewed to implement the many features the suite now offers, such as the different parameters, modes, and the user interface. I take full responsibility for the resulting code.

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List of Symbols

Bold symbols denote vectors or matrices, shown in vector notation even where the two-terminal case reduces them to scalars.

Notation	Description	Page List
$\mathbf{v}_C$	Circuit-side port voltage	4, 6, 7, 10
$\mathbf{v}_F$	Field-side port voltage	6–8, 10
$\mathbf{i}_C$	Circuit-side port current	4, 6, 7, 10, 17, 24
$\mathbf{i}_F$	Field-side port current	6–8, 10
$\mathbf{M}$	Modified Nodal Analysis “mass” matrix	4, 7, 8
$\mathbf{K}$	Modified Nodal Analysis “stiffness” matrix	4, 7, 8
$\mathbf{Z}$	Port impedance of the field subsystem	7, 8
$\mathbf{Y}$	Port admittance, $\mathbf{Y} = \mathbf{Z}^{-1}$	7, 8
$\mathbf{L}$	Port inductance, $\mathbf{Z} \approx j\omega\mathbf{L}$ at low frequencies	6, 8–10, 14
$\mathbf{R}$	Port resistance	8–10, 14
T_n	n -th synchronisation point, window end	5, 10, 11, 15, 24
H_n	Window size $T_{n+1} - T_n$	5
ρ	Contraction factor	7
β	Weighting factor in the generalised condition	7
ω	Angular frequency	7, 8

1 Introduction

1.1 Motivation

Most parts of a drive system, such as sources, switches, passive elements, and control circuitry, do not need to be modelled with the full spatial detail of Maxwell’s partial differential equations [16] and can be adequately described by lumped elements. These reduced models are formulated as differential-algebraic equations (DAEs), e.g. by modified nodal analysis (MNA) [8]. For the machine itself, a lumped representation may be insufficient whenever saturation, eddy-current effects or the spatial distribution of the field govern its behaviour. A spatially distributed model is used instead, commonly obtained from the magnetoquasistatic (MQS) approximation of Maxwell’s equations and discretised in space, e.g. by the finite element method (FEM) [10]. Coupling such a field model to the circuit yields a system of partial differential-algebraic equations (PDAEs), which reduces to a system of DAEs once the field model is discretised in space [2]. Solving the system *monolithically* would require a single time integrator to resolve the dynamics of the most active domain, so the latent domain is advanced with a step size far smaller than its dynamics require. Partitioning removes this restriction, as each subsystem can be integrated with a step size adapted to its own time constants. This is particularly beneficial when the time scales of the domains are separated by orders of magnitude, where it has been shown to reduce the computational effort considerably [15]. Electric machines are increasingly fed by fast-switching power electronics, enabled by wide-bandgap semiconductor devices and driven by “ambitious” power-density and voltage targets for the coming decade, see Tab. 1.1. In such systems, the machine is the latent subsystem, as its electromagnetic and mechanical time constants exceed the converter switching period by orders of magnitude, which motivates multirate and co-simulation approaches [14, 11]. Faster switching also increases the voltage slew rate dv/dt at the machine terminals. Whereas insulation systems of standard machines are designed for slew rates of a few $kV\mu s^{-1}$, silicon-carbide devices reach values exceeding $100kV\mu s^{-1}$, according to [12]. Consequently, the behaviour of the machine at its terminals is no longer adequately represented by a lumped element, motivating a partitioned field-circuit approach in which the interface conditions capture the machine’s terminal response.

Practically speaking, there are well-established solvers for each domain. However, commercial software is often closed-source and access to the internal variables during simulation is limited, making coupling the field-circuit domains difficult [7]. Hence, a so-called *black-box* view of the problem, which is agnostic to the solvers, is the versatile implementation. More precisely, one might call it a *grey-box* view, since we require some knowledge of the state of the field domain in the interface condition, as we will discuss later.

1.2 State of the Art

Solving the system of DAEs is a thoroughly studied problem in the context of simulating field-circuit domains. Approaches differ in how tightly the subsystems are coupled, ranging from monolithic solves, where all

Table 1.1: Power electronics and electric motor targets (US DoE).

	2025	2030	2035
Voltage (V)	600	800	800
Power Density Electronics (kW/L)	150	200	225
Power Density Machines (kW/L)	75	100	112.5

Data from [17].

equations are assembled into a single system, to loosely coupled *dynamic iteration* (also called *Waveform Relaxation* (WR)) where the subsystems are solved separately and exchange waveforms over time windows, see Section 2.2. The monolithic approach requires a joint solver for both domains, which conflicts with the black-box setting and is prohibitively expensive in computation for the targeted application here, see Section 1.1. In the case of dynamic iteration, it has been identified that the exchange of algebraic variables can be a mathematical cause for divergence [14]. Thus, enhanced modelling of the interface via a *surrogate* or *reduced order model* (ROM) has been shown to improve convergence significantly. Without it, the iteration may be divergent [1]. While such an interface condition is straightforward to formulate analytically, its time discretisation involves the time integrators of both subsystems. When black-box tools are coupled, each subsystem discretises the shared condition with its own integrator, so the two discrete operators may differ. Different operators result in an inconsistent interface condition and may possibly lead to a non-converging iteration, as we will see in Chapter 5. This motivates the need for a consistent implementation of the interface without access to the solver’s time integrator, see Chapter 3. Xyce™ also provides a native coupling interface (General External Interface). Why it is not used for the dynamic iteration here is discussed in Section 4.4.

1.3 Objective and Contributions

The contributions of this thesis are:

1. a dynamic iteration co-simulation of the field-circuit built entirely from black-box, netlist-driven simulators (demonstrated with Xyce™);
2. a consistent formulation and implementation of the interface condition that does not require access to either solver’s time integrator;
3. a study of the resulting scheme’s convergence behaviour, transmission conditions and parameter-dependence, e.g. window-length, isolated from any particular field discretisation.

Xyce™ is an open-source analogue circuit simulator used in this thesis. It can be replaced by other netlist-driven simulators with little change to the implementation, see Section 4.1.

1.4 Outline of the Thesis

In Chapter 2, the fundamental theory that is required to explain the implementation and discuss any results is laid out. In Chapter 3 we analyse different discretisations of the interface condition. Chapter 4 documents the implementation of the co-simulation with the Xyce™ parallel electronic simulator. In Chapter 5, the



implementation is verified against a suite of benchmark problems. Chapter 6 contains parameter studies and a general approach to choosing them. Chapter 7 concludes the thesis and outlines possibilities for future work.

2 Fundamentals

Decomposing large systems into coupled subsystems that can be integrated independently is a key strategy that can improve simulation speed by up to two orders of magnitude [7]. This approach, known as *domain decomposition*, follows the principle of *divide et impera* (Latin for “divide and conquer”). Before the coupling between the field and circuit domains can be realised through dynamic iteration, the underlying concepts must first be introduced.

2.1 Modified Nodal Analysis

The circuit subsystem can be modelled as a directed graph with an incidence matrix $\mathbf{A}$ that dictates the layout of the lumped circuit elements. KIRCHHOFF’s circuit laws form the foundation of lumped-element network analysis. In matrix form, they are written as

$$\mathbf{A}\mathbf{i}(t) = 0 \quad \text{and} \quad \mathbf{u}(t) = \mathbf{A}^\top \varphi(t),$$

where $\mathbf{i}$ is the vector of branch currents and φ is the vector of node potentials. Based on these rules, a system of DAEs that governs the behaviour of the circuit subsystem can be written in abstract form¹ as

$$\mathbf{M} \frac{d}{dt} \mathbf{x}(t) + \mathbf{K} \mathbf{x}(t) = \mathbf{r}(t)$$

where, in the case of MNA, $\mathbf{x}$ denotes the unknown functions of node potentials and the voltage-source and inductor branch currents

$$\mathbf{x}^\top(t) = [\varphi^\top(t) \quad \mathbf{i}_L^\top(t) \quad \mathbf{i}_V^\top(t)].$$

Moreover, we introduce

$$\mathbf{v}_C(t) = \mathbf{P}^\top \mathbf{x}(t) \quad \text{and} \quad \mathbf{P} \mathbf{i}_C(t),$$

where $\mathbf{v}_C$ is the voltage and $\mathbf{i}_C$ the current at the field-circuit interface, and $\mathbf{P}$ is the corresponding port operator. Using this notation, the (black box) field subsystem may be abstractly written as

$$\frac{d}{dt} \mathbf{x} = f(\mathbf{x}, t, \mathbf{i}_C).$$

¹Throughout this document, we retain generic vector/matrix notation even where the two-terminal (single port) scenarios reduce the port quantities to scalars. This is done in anticipation of multiport generalisation.

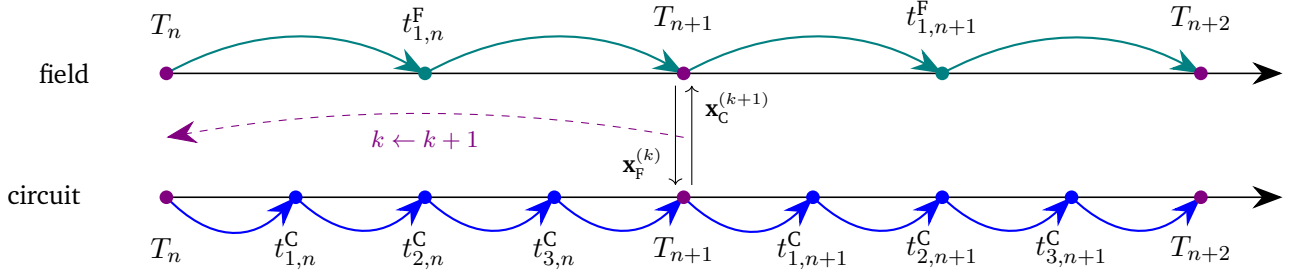


Figure 2.1: Dynamic iteration. The two subsystems (field/circuit) are evaluated on their own time grids, with different step sizes (solid teal and blue arrows). A window is the time span between synchronisation points T_n (purple dots). Only at a synchronisation point are the state vectors updated, in this case according to a Gauss–Seidel scheme. The solution is computed iteratively on a window and uses rollback (dashed purple arrow).

2.2 Dynamic Iteration

The subsystems are integrated independently for a period of time $\mathcal{T}_n \rightarrow \mathcal{T}_{n+1}$, where $\mathcal{T}_n = [T_n, T_{n+1}]$ with window size $H_n := T_{n+1} - T_n$, using the outputs of neighbouring subsystems as inputs; see Figure 2.1. This is the essence of dynamic iteration [7]. It is a fixed-point iteration technique, i.e., given a function $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$, the iteration

$$\mathbf{x}^{(k+1)} = \phi(\mathbf{x}^{(k)}), \quad k = 0, 1, 2, \dots$$

converges to a fixed point, where $\mathbf{x}^* = \phi(\mathbf{x}^*)$, [4].

In the case of a field-circuit coupling, the vector of unknowns $\mathbf{x}$ contains the voltage (node potentials) and (branch) current across the interface ports. The fixed point is reached if the voltages and currents of both subsystems converge, i.e.,

$$\underbrace{\begin{matrix} \mathbf{v}_C^{(k+1)} = \mathbf{v}_C^{(k)} \\ \mathbf{i}_C^{(k+1)} = \mathbf{i}_C^{(k)} \end{matrix}}_{\text{circuit}} \quad \underbrace{\begin{matrix} \mathbf{v}_F^{(k+1)} = \mathbf{v}_F^{(k)} \\ \mathbf{i}_F^{(k+1)} = \mathbf{i}_F^{(k)} \end{matrix}}_{\text{field}}$$

where $\mathbf{v}_C$ and $\mathbf{i}_C$ are the port voltage and current of the circuit-side, and $\mathbf{v}_F$ and $\mathbf{i}_F$ are the port voltage and current of the field-side.

2.3 Coupling Scheme

Transmission between the subsystems is handled at the ports, i.e., the inputs and outputs of a subsystem. Two independent factors determine field-circuit coupling: the coupling direction and the coupling conditions themselves.

2.3.1 Coupling Direction

The direction of coupling stipulates which variable from one subsystem (output) is prescribed on another (input) during iteration. For our field-circuit co-simulation, it answers the question: *“Does the field ingest a current or a voltage as its (plain) input?”* This consideration directly impacts the index of the resulting DAE.

Roughly speaking, the index of a DAE corresponds to the highest derivative of an input function that enters the subproblem [14].

Example 2.1 (DAE index). Take the current-driven case, where the circuit’s port-current $\mathbf{i}_F(t) = \mathbf{i}_C(t)$ is prescribed as the input function on the field subsystem, thereby *driving* it: The variable to solve for is the field’s port-voltage $\mathbf{v}_F$, which, in a predominantly inductive problem is given by

$$\mathbf{v}_F(t) \approx \mathbf{L} \frac{d}{dt} \mathbf{i}_F(t).$$

The input function requires one differentiation here. If we look at the voltage-driven case on the other hand, the field solves for

$$\mathbf{i}_F(t) \approx \mathbf{i}_{F,0} + \mathbf{L}^{-1} \int_0^t \mathbf{v}_F(\tau) d\tau,$$

requiring no differentiation, but an integration instead, which actually has a smoothing property. Based on this simplistic example, one might easily understand that a lower-index system is desirable, since numerical differentiation amplifies noise and discretisation error.

The example above ignores the algebraic constraints of the full coupled DAE. In the case of more general systems, it has been shown that voltage-driven field formulations result in a lower-index system (index-1) and thereby offer better numerical stability than the current-driven alternative (index-2) [2]. Hence, a voltage-driven coupling

$$\mathbf{v}_F \leftarrow \mathbf{v}_C$$

is likely the preferred choice for this work’s implementation; see Chapter 4, though under some scopes the current-driven approach may be considered, cf. e.g. [3].

2.3.2 Coupling Conditions

The coupling conditions, also called transmission conditions in the context of SCHWARZ coupling, specify *how* the exchanged variables enter each subsystem as inputs and leave it as outputs, see e.g. [6]. For a voltage-driven coupling, the plain (zero-order) conditions of the field-circuit problem read

$$\mathbf{v}_F \leftarrow \mathbf{v}_C \quad \text{and} \quad \mathbf{i}_C \leftarrow \mathbf{i}_F, \tag{2.1}$$

that is, the field is prescribed the voltage and returns the current. The exchanged quantities are waveforms, $\mathbf{v}_C = \mathbf{v}_C(t)$ and $\mathbf{i}_F = \mathbf{i}_F(t)$, which gives the method its name.

Extracting a reduced-order model (ROM) of the field domain improves the coupling, and the better the ROM, the larger the admissible time windows [15, 14]. Because the spatially discretised magnetoquasistatic field behaves as an inductance-like element, the resulting ROM is itself inductive [2]. Later work established that this inductance is precisely the first-order term of a Neumann-series expansion of the optimal transmission condition [3].

Following the suggestion of [3] to excite the field by voltages rather than currents, the optimal transmission condition follows by an analogous derivation.

Let the field problem be voltage-driven within a GAUSS–SEIDEL scheme. The zero-order Dirichlet conditions read

$$\mathbf{v}_F^{(k+1)} = \mathbf{v}_C^{(k+1)}, \quad (2.2a)$$

$$\mathbf{i}_C^{(k+1)} = \mathbf{i}_F^{(k)}. \quad (2.2b)$$

The circuit's input condition (2.2b) is generalised to a linear combination of current and voltage,

$$\mathbf{i}_C^{(k+1)} - \beta \mathbf{v}_C^{(k+1)} = \mathbf{i}_F^{(k)} - \beta \mathbf{v}_F^{(k)}, \quad (2.3)$$

where β is a weighting factor. Only the circuit's input condition is generalised. The field's condition (2.2a) is left plain. The field admits an easily extractable impedance, but an arbitrary circuit does not, so optimising the field-side condition analogously is cumbersome [3].

Eliminating the internal degrees of freedom from the frequency-domain circuit equations yields the port relation $\mathbf{v}_C = \mathbf{r}_C - \mathbf{x}_P \mathbf{i}_C$, with transfer impedance $\mathbf{x}_P(\omega) = \mathbf{P}^\top (\mathbf{M} j\omega + \mathbf{K})^{-1} \mathbf{P}$ and projected source term $\mathbf{r}_C = \mathbf{P}^\top (\mathbf{M} j\omega + \mathbf{K})^{-1} \mathbf{r}(\omega)$. The voltage-driven field returns $\mathbf{i}_F = \mathbf{Y} \mathbf{v}_F$, where the admittance $\mathbf{Y}(\omega) = \mathbf{Z}^{-1}(\omega)$ is the inverse of the port impedance $\mathbf{Z}(\omega)$ of the MQS problem [3]. Inserting the field response and the plain condition (2.2a) into (2.3) and eliminating $\mathbf{v}_C^{(k+1)}$ gives the error propagation

$$\mathbf{i}_C^{(k+1)} - \mathbf{i}_C^{(k)} = \rho(\beta) (\mathbf{i}_C^{(k)} - \mathbf{i}_C^{(k-1)}), \quad \rho(\beta) = (\mathbf{I} + \beta \mathbf{x}_P)^{-1} (\beta - \mathbf{Y}) \mathbf{x}_P,$$

where $\mathbf{I}$ is the identity matrix.

In order for the fixed point iteration to converge, the error across iterations has to contract, i.e., $|\rho(\beta)| < 1$. Assuming $\mathbf{x}_P$ is invertible, optimal convergence $\rho(\beta) = 0$ is attained for

$$\beta = \mathbf{Y}(\omega),$$

so that the optimised circuit-side transmission condition reads

$$\mathbf{i}_C^{(k+1)} - \mathbf{Y}(\omega) \mathbf{v}_C^{(k+1)} = \mathbf{i}_F^{(k)} - \mathbf{Y}(\omega) \mathbf{v}_F^{(k)}, \quad (2.4)$$

which is equivalent to the transmission condition

$$\mathbf{v}_C^{(k+1)} = \mathbf{Z}(\omega) \mathbf{i}_C^{(k+1)} + \mathbf{v}_F^{(k)} - \mathbf{Z}(\omega) \mathbf{i}_F^{(k)} \quad (2.5)$$

derived from the current-driven setting in [3]. This leads to the following result.

Result 2.1. *The optimisation of the circuit-side transmission condition is impartial to the input function of the field subsystem, i.e., imposing $\mathbf{v}_C$ results in the same optimal transmission condition (2.5) as imposing $\mathbf{i}_C$.*

They can be viewed as Thévenin and Norton equivalents, cf. [5]. This is expected, as both port variables appear on each side of either optimised transmission condition, tied together by the exact port operator ($\mathbf{Z}$ or $\mathbf{Y}$). That operator reproduces the field's response exactly and, trivially, would guarantee convergence of the scheme in one iteration ($\rho = 0$).

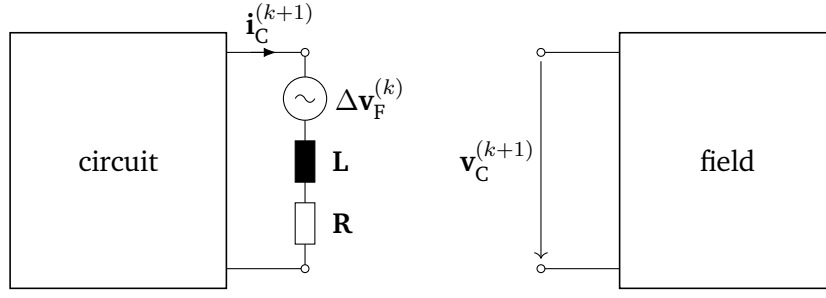


Figure 2.2: Schematic representation of the $k + 1$ -th Gauss–Seidel iteration for the field-circuit subsystems, using an approximation of the optimal transmission coupling condition.

Since $\mathbf{Z} \approx j\omega\mathbf{L}$ is the low-frequency ($\omega\tau \ll 1$) approximation of the exact port impedance (the leading term of the Neumann series, corresponding to the magnetostatic inductance; see [3]), the admittance inherits $\mathbf{Y} \approx (j\omega\mathbf{L})^{-1}$ in the same regime. As in the current-driven case, we must approximate this frequency-domain expression in our time-domain implementation, where $j\omega$ corresponds to a differentiation in time [3]. The resulting condition

$$\mathbf{i}_F(t) \approx \mathbf{L}^{-1} \int \mathbf{v}_F(t) dt$$

represents the dual of the flux-linkage condition

$$\mathbf{v}_F(t) \approx \mathbf{L} \frac{d}{dt} \mathbf{i}_F(t)$$

of the current-driven scheme, where $\mathbf{L}$ is constant. Since the underlying inductance is relatively cheap to extract (a single magnetostatic solve), it could be re-evaluated at successive working points throughout the transient to track the magnetic saturation effect of the field subsystem, at the cost of repeated extraction; this is not implemented in this work. Eddy-current effects are not covered in this first-order approximation of the inductance; those dynamics are compensated by the lagged correction term $\Delta \mathbf{i}_F^{(k)} := \mathbf{i}_F^{(k)} - \mathbf{Y}(\omega) \mathbf{v}_F^{(k)}$ in Equation (2.4) or $\Delta \mathbf{v}_F^{(k)} := \mathbf{v}_F^{(k)} - \mathbf{Z}(\omega) \mathbf{i}_F^{(k)}$ in Equation (2.5), respectively.

The impedance of our ROM is complemented by a resistance $\mathbf{R}$, which may also be extracted by a port measurement of the subsystem². Hence, expanding the low-frequency approximation of transmission condition (2.5) to

$$\mathbf{v}_C^{(k+1)} = \mathbf{v}_F^{(k)} + \left(\mathbf{R} + \mathbf{L} \frac{d}{dt} \right) (\mathbf{i}_C^{(k+1)} - \mathbf{i}_F^{(k)})$$

in the time domain, see Figure 2.2.

Algorithm With this information, an algorithm in the time domain can be written as follows.

$$\mathbf{M} \frac{d}{dt} \mathbf{x}^{(k+1)} + \mathbf{K} \mathbf{x}^{(k+1)} + \mathbf{P} \mathbf{i}_C^{(k+1)} = \mathbf{r}(t) \quad (2.6)$$

$$\mathbf{v}_C^{(k+1)} = \mathbf{P}^\top \mathbf{x}^{(k+1)} \quad (2.7)$$

²The resistance can be calculated according to OHM's law, e.g., by imposing a DC current of 1 A at the terminals and measuring the resulting potential difference.

with the coupling condition

$$\mathbf{v}_C^{(k+1)}(t) = (\mathbf{R} + \mathbf{L} \mathbf{d}_t) \mathbf{i}_C^{(k+1)}(t) + \Delta \mathbf{v}_F^{(k)}(t) \quad (2.8)$$

$$\Delta \mathbf{v}_F^{(k)}(t) := \mathbf{v}_F^{(k)}(t) - (\mathbf{R} + \mathbf{L} \mathbf{d}_t) \mathbf{i}_F^{(k)}(t) \quad (2.9)$$

Step 0. Set $k = 0$ and $\Delta \mathbf{v}_F^{(0)}(t) = \Delta \mathbf{v}_{F,0}(t)$, and extract $\mathbf{L}$.

Step 1. Solve the circuit system with condition (2.8) for the given input $\Delta \mathbf{v}_F^{(k)}(t)$ to obtain the output $\mathbf{v}_C^{(k+1)}(t)$.

Step 2. Solve the field system for the given input $\mathbf{v}_C^{(k+1)}(t)$ to obtain $\mathbf{i}_F^{(k+1)}(t)$, and update the correction $\Delta \mathbf{v}_F^{(k+1)}(t)$ according to (2.9).

Step 3. If not converged, set $k = k + 1$ and go to Step 1. Otherwise proceed with $\Delta \mathbf{v}_F^{(0)}(t) = \Delta \mathbf{v}_F^{(k+1)}(t_{n+1})$ for $t \in T_{n+1}$, set $n = n + 1$ and go to Step 1.

Note on Solving the DAEs Though we used a GAUSS–SEIDEL scheme (Einzelschrittverfahren) to obtain a solution of the DAEs in this implementation, other algorithms such as GAUSS–JACOBI (Gesamtschrittverfahren) are common, too. While the former offers faster convergence, the latter is more parallelisable [1].

3 Interface Discretisation

The interface condition from Chapter 2 is time-continuous. Coupling black-box tools means each solver discretises the condition with its own integrator, so the two discrete operators may differ. In this chapter we analyse the discretisation error and show how an accumulated-derivative formulation removes part of it at the synchronisation points.

If the Scheme (2.6)–(2.9) converges, i.e.,

$$\mathbf{v}_C(t) = \mathbf{v}_F(t), \quad \mathbf{i}_C(t) = \mathbf{i}_F(t) \quad \text{and} \quad \mathbf{v}_C^{(k+1)}(t) = \mathbf{v}_C^{(k)}(t), \quad \mathbf{i}_F^{(k+1)}(t) = \mathbf{i}_F^{(k)}(t)$$

the terms in the time-continuous formulation of the interface condition (2.8)–(2.9) cancel out:

$$\mathbf{v}_C^{(k+1)}(t) = \cancel{(\mathbf{R} + \mathbf{L} \mathbf{d}_t)} \mathbf{i}_C^{(k+1)}(t) + \mathbf{v}_F^{(k)}(t) - \cancel{(\mathbf{R} + \mathbf{L} \mathbf{d}_t)} \mathbf{i}_F^{(k)}(t),$$

whereby the equations are solved consistently. Consistency cannot be guaranteed, however, when the transmission condition is discretised in time.

For a given window $I_j = [t_0, t_0 + H_j]$ with $t_0 = T_n$, the different solvers start from the same initial value, but may integrate on different time grids thereafter; see Figure 2.1.

$$\begin{aligned} \text{circuit:} \quad & t_0 < t_1^C < t_2^C < \dots < t_0 + H_j, & h_C &:= t_p^C - t_{p-1}^C, \\ \text{field:} \quad & t_0 < t_1^F < t_2^F < \dots < t_0 + H_j, & h_F &:= t_r^F - t_{r-1}^F. \end{aligned}$$

Hence, applying backward EULER yields

$$\begin{aligned} \mathbf{v}_C^{(k+1)}(t_p^C) &= \mathbf{R} \mathbf{i}_C^{(k+1)}(t_p^C) + \mathbf{L} \frac{\mathbf{i}_C^{(k+1)}(t_p^C) - \mathbf{i}_C^{(k+1)}(t_{p-1}^C)}{h_C} + \Delta \mathbf{v}_F^{(k)}(t_p^C), \\ \Delta \mathbf{v}_F^{(k)}(t_r^F) &= \mathbf{v}_F^{(k)}(t_r^F) - \mathbf{R} \mathbf{i}_F^{(k)}(t_r^F) - \mathbf{L} \frac{\mathbf{i}_F^{(k)}(t_r^F) - \mathbf{i}_F^{(k)}(t_{r-1}^F)}{h_F}. \end{aligned}$$

The circuit needs $\Delta \mathbf{v}_F^{(k)}(t_p^C)$, but t_p^C is a circuit node and is (in general) not a field node. Hence, an interpolation of the two field samples that bracket the circuit node is needed. A linear interpolation yields

$$\Delta \mathbf{v}_F^{(k)}(t_p^C) = \Delta \mathbf{v}_F^{(k)}(t_{r-1}^F) + \frac{t_p^C - t_{r-1}^F}{t_r^F - t_{r-1}^F} (\Delta \mathbf{v}_F^{(k)}(t_r^F) - \Delta \mathbf{v}_F^{(k)}(t_{r-1}^F)), \quad t_{r-1}^F \leq t_p^C \leq t_r^F.$$

Then, if the scheme converges, i.e., $\mathbf{v}_C(t_p^C) = \mathbf{v}_F(t_p^C)$ and $\mathbf{i}_C(t_p^C) = \mathbf{i}_F(t_p^C)$ and with no iteration change, the terms cancel and consistency is guaranteed.

Result 3.1. A “naive” discretisation of the field-circuit transmission coupling condition (2.8)–(2.9), evaluated on the circuit, requires interpolation of the field quantities whenever the subsystems do not share a common grid. Provided the scheme converges, the equations are then solved consistently up to the discretisation error of the underlying method—in our case $\mathcal{O}(h_C) + \mathcal{O}(h_F)$ for backward Euler.

Since both subsystems restart from the same initial current at t_0 , using $\mathbf{i}_C^{(k+1)}(t_0) = \mathbf{i}_F^{(k)}(t_0)$, $\forall k$ we propose a different formulation of the interface condition (2.8)–(2.9), by taking the *accumulated* derivative

$$\Delta d_t \mathbf{i}(t) = \frac{\mathbf{i}_C^{(k+1)}(t) - \mathbf{i}_C^{(k+1)}(t_0)}{t - t_0} - \frac{\mathbf{i}_F^{(k)}(t) - \mathbf{i}_F^{(k)}(t_0)}{t - t_0} = \frac{\mathbf{i}_C^{(k+1)}(t) - \mathbf{i}_F^{(k)}(t)}{t - t_0}, \quad t \in [t_0, t_0 + H_j], \quad (3.1)$$

in both subsystems. Equation (3.1) is a global (window) secant, anchored at t_0 .

Result 3.2. *Using the accumulated (secant) derivative (3.1) in the interface condition (2.8)–(2.9) and interpolating the lagged field correction $\Delta \mathbf{v}_F^{(k)}$, the discrete residual $\Delta d_t \mathbf{i}(t)$ vanishes at convergence. Because of the anchored denominator $t - t_0$, there is a roll-off as t increases for a given window.*

Both methods can be used to discretise the interface condition, as we will see later. Because of their different behaviour, however, especially regarding the time between synchronisation points, it seems advisable to use different convergence criteria. While the backward EULER condition permits many different criteria, such as the relative iteration change

$$\frac{\int_{T_n}^{T_{n+1}} |\mathbf{i}_F^{(k)} - \mathbf{i}_F^{(k-1)}| dt}{\int_{T_n}^{T_{n+1}} |\mathbf{i}_F^{(k)}| dt} \leq \text{Tol}, \quad (3.2)$$

cf. [3], we found the accumulated condition is only usable with a terminal criterion, which evaluates the exchanged variables at the synchronisation points T_n , i.e.,

$$|\mathbf{i}_C^{(k+1)}(T_n) - \mathbf{i}_C^{(k)}(T_n)| + |\mathbf{v}_C^{(k+1)}(T_n) - \mathbf{v}_C^{(k)}(T_n)| \leq \text{Tol} \quad \text{and} \quad \begin{cases} |\mathbf{i}_C^{(k+1)}(T_n) - \mathbf{i}_F^{(k+1)}(T_n)| \leq \text{Tol} & \text{voltage-driven,} \\ |\mathbf{v}_C^{(k+1)}(T_n) - \mathbf{v}_F^{(k+1)}(T_n)| \leq \text{Tol} & \text{current-driven.} \end{cases}$$

The accumulated derivative is an anchored secant of the current state change from the window start. It does not approximate the instantaneous derivative in the window interior. Besides the discretisation error, Equation (3.1) also makes it easy to see that any errors or noise in the current are amplified near the window start as $t \rightarrow t_0$. Using the terminal criterion then causes a relatively large error between field and circuit currents at the window starts, which manifests as voltage “spikes” (sometimes also ramps) in our simulations. In our experience however, even a global criterion such as (3.2) will result in such spikes, provided a small enough step by the circuit’s integrator.

4 Coupling with Xyce™

This chapter discusses the implementation of the dynamic iteration scheme presented in Chapters 2 and 3. At this point in the thesis, one may rightfully ask *Why Xyce™*? The answer is that we do not specifically need it: although Xyce™ was used here as the solver for the circuit subsystem, one can define general requirements that allow it to be replaced by other solvers.

4.1 Requirements on the Circuit Solver

To implement dynamic iteration, which is a fixed-point iteration technique, a rollback that repeats the co-simulation step with corrected inputs is required [7]. In Xyce™, this rollback is called *restarting* and is accomplished by *checkpointing*, whereby the completed simulation state is saved periodically [9]. The second requirement on the solver is what Xyce™ aptly calls *Analog Behavioural Modeling* (ABM), which allows a circuit element to be described by a mathematical relationship [9]. This feature is needed to implement one of the interface conditions derived in Chapter 3.

4.2 Field Dummy Solver

For the field subsystem, we decided that the extensive capabilities of an established solver were not necessary. One of the goals defined in Section 1.3 was to implement the co-simulation in a manner that is agnostic to the choice of solver. We therefore opted for a simplistic “dummy” solver that emulates a device with plausible physical behaviour.

We assigned the subsystem its own inductance and resistance values – treated as the *true* values for our purposes – which may differ from those the interface condition uses as the ROM. There are two solver-methods, one for each coupling direction, and the C++ coupling driver calls them from the same file. Xyce’s output is first resampled onto a uniform grid of `N_field_eval_intervals` points, on which the field solver computes either the current or the voltage, depending on the drive direction. In the voltage-driven case, the method `FEM_solver_voltage_driven_waveform` computes the current via

```
double dt = V_eval.t[j] - V_eval.t[j - 1];  
double V_p = V_eval.y[j]; // voltage at the j-th sample  
double I_field = (V_p + L_FEM * I_ref / dt) / (R_FEM + L_FEM / dt);
```

at each point of the resampled waveform. In other words, this yields the new current for the given terminal voltage using a backward EULER discretisation of the current derivative. Analogously, the current-driven solver returns the field voltage according to

```
double dt = I_eval.t[j] - I_eval.t[j - 1];  
double dl_dt = L_FEM * (Ib - Ia) / dt; // differentiated flux (backward euler)  
double V_field = R_FEM * I_eval.y[j] + dl_dt;
```

This approach has the obvious advantage of being very fast to compute, at the cost of leaving the implementation untested against a full 3D field solver.

4.3 Coupling via the Xyce Binary

With the requirements from Section 4.1, coupling to Xyce™ can be obtained purely through netlists and without the need to modify and build from the source code.¹ This section explores how Xyce™ implements the requirements from above.

4.3.1 Restarts and Checkpointing

When “checkpointing” is used, Xyce periodically saves its completed simulation state. The saved state can be used to restart Xyce™ from one of these “checkpoints”. This feature, originally intended as a backup for long running simulations, can be used to implement our rollback requirement. Xyce uses the `.OPTIONS RESTART` netlist command to control all checkpoint output and restarting [9]. We use

```
.OPTIONS RESTART PACK=0 JOB=ckpt_out INITIAL_INTERVAL={dt_window}
```

as the restart directive on the first window. `PACK=0` indicates that restart data files contain ASCII data, instead of binary (`PACK=1`). `JOB` determines the suffix of the file name the data is saved in and `INITIAL_INTERVAL` identifies the save interval (we only need the window end). Naturally, we use the transient directive

```
.tran {dt_print} {t_stop} {t_abs_start} UIC
```

to simulate the circuit on the first window, where `dt_print` is the minimum interval in which Xyce™ must output its solution, `t_stop` is the time at the window end, `t_abs_start` is the starting time ($t = 0$ on the first window) and `UIC` is only used on the first window and tells the solver to use any initial conditions provided in the netlist.

On any other window after the first, we use

```
.OPTIONS RESTART FILE={comitted_file} JOB=ckpt_out INITIAL_INTERVAL={dt_window}  
.tran {dt_print} {t_stop} {t_abs_start}
```

as directives, where `FILE` specifies the name of the file from which to restart from (rollback to).

¹While we built it from source, the developers of Xyce™, Sandia National Laboratories, also provide executables for MacOS, Microsoft Windows, and RedHat Enterprise. Note that, as of the writing of this thesis, the developers publish executables based on the “XyceNF” version of the code, rather than the open-source version, and they may stop doing so entirely in the future.

4.3.2 Interface Condition via Behavioural Source

To implement the interface condition, we use the ABM capability of Xyce™. ABM devices (B devices) are specified in a netlist the same way as other devices. They can serve as either a voltage or current source, meaning an interface condition can be expressed as $\mathbf{v}_C^{(k+1)}$ or $\mathbf{i}_C^{(k+1)}$ from our GAUSS–SEIDEL scheme. We customise the operational behaviour of the B device via

```
VFprev vfprev 0 PWL FILE "vf_prev_k.pwl"
VIprev iprev 0 PWL FILE "i_prev_k.pwl"
Vmeas p nx 0
Bfield nx 0 V = { V(vfprev) + (Rrom + Lrom/(time - t_abs_start)) * (I(Vmeas) - V(iprev)) }
```

to represent the accumulated interface condition derived in Chapter (3), where Bfield equates with $\mathbf{v}_C^{(k+1)}$. Here, time is a keyword reserved by Xyce™ and used by the solver's integrator to evaluate the current time in the simulation [9]. $I(Vmeas)$ is equivalent to $\mathbf{i}_C^{k+1}$ and acts as an ideal amperemeter (no voltage or resistance) in series with the interface. $V(vfprev)$ and $V(iprev)$ are the piecewise linear (PWL) sources that contain the field solution of the previous iteration. They are encoded as voltage sources VFprev, VIprev and not connected to any other nodes of the circuit. This is standard SPICE idiom, because a voltage source allows a direct lookup, $V(node)$, as apposed to encoding them in an isolated current source, which would yield a singular KCL row in the MNA system. Lrom and Rrom are the inductance and resistance values used for the ROM.

We also implemented the “naive” backward EULER discretisation of the interface, where Lrom and Rrom are ideal circuit elements in the netlist and are no longer present in the B source. We modelled the interface via

```
Vdidtf vdidtf 0 PWL FILE "didt_field_k.pwl"
Rromc nx nrc {Rrom}
Lromc nrc nlc {Lrom} IC={I0}
Bfield nlc 0 V = { V(vfprev) - Rrom*V(iprev) - Lrom*V(vdidtf) }
```

where $V(vfprev)$ and $V(iprev)$ remain the same as before. Vdidtf introduces the backward EULER discretisation of the field subsystem and is calculated by the dummy-solver, interpolated as PWL. Here, we only dictate $\Delta \mathbf{v}_F^{(k)}$ and Xyce™ handles the computation of

$$\mathbf{R} \mathbf{i}_C^{(k+1)}(t_p^C) + \mathbf{L} d_t \mathbf{i}_C^{(k+1)}$$

on its own grid. Besides backward EULER (first-order trapezoidal method), Xyce™ actually allows users to specify the time integration method with an .OPTIONS line in the netlist to methods of higher order [9].

Furthermore, to determine VFprev, VIprev, and Vdidtf, we used linear extrapolation at the first iteration of a given window. The slope of that linear ramp is determined by the last two points of the converged solution of the previous window.

4.3.3 Coupling Direction

As explained in Section 2.3, the circuit subsystem can impose either a voltage or a current on the terminals of the field subsystem. A current-driven formulation, i.e., $\mathbf{i}_F^{(k+1)}(t) = \mathbf{i}_C^{(k+1)}(t)$, changes the meaning of the PWL sources VFprev, VIprev and Vdidtf from Subsection 4.3.2². The variables are compared in Table 4.1.

²Note that the coupling direction has nothing to do with the source type that a given interface condition is expressed in on the circuit-side.

Table 4.1: Comparison of interface condition variables depending on coupling direction.

	voltage-driven	current-driven
VFprev	$\mathbf{v}_F^{(k)}(t) = \mathbf{v}_C^{(k)}(t)$	$\mathbf{v}_F^{(k)}(t)$
VIprev	$\mathbf{i}_F^{(k)}(t)$	$\mathbf{i}_F^{(k)}(t) = \mathbf{i}_C^{(k)}(t)$
Vdidtf	$\mathbf{d}_t \mathbf{i}_F^{(k)}(t)$	$\mathbf{d}_t \mathbf{i}_F^{(k)}(t) = \mathbf{d}_t \mathbf{i}_C^{(k)}(t)$

As was previously established, voltage-driven field formulations result in a lower-index system and thereby offer better numerical stability than the current-driven alternative. It is worth noting however, that a current-driven formulation also impacts the meaning of the time derivatives in the interface conditions we derived. A current-driven model, where $\mathbf{i}_F^{(k)}(t) = \mathbf{i}_C^{(k)}(t)$, results in an iteration-difference of the same derivative, i.e.,

$$\Delta \mathbf{d}_t \mathbf{i}(t) = \mathbf{d}_t \mathbf{i}_C^{(k+1)}(t) - \mathbf{d}_t \mathbf{i}_C^{(k)}(t),$$

instead of a cross-solver difference

$$\Delta \mathbf{d}_t \mathbf{i}(t) = \mathbf{d}_t \mathbf{i}_C^{(k+1)}(t) - \mathbf{d}_t \mathbf{i}_F^{(k)}(t),$$

from the familiar voltage-driven model. While this may have little impact on a converged solution using a global criterion, e.g. the L1 norm from Equation (3.2), this more severely impacts the window interiors of the accumulated interface condition using the terminal criterion. There, the cross-solver difference may be much larger than any iterative difference near the window start. Hence, when current-driven modelling is viable, this may reduce the previously mentioned (voltage) spikes.

4.4 General External Interface and Its Limitations

The General External Interface is an API developed by Xyce™ to couple a circuit to a specialised domain simulation run by another code. This allows users to construct a circuit netlist that includes both real circuit elements and special general external devices (GENEXT devices for short) which serve as opaque coupling devices that invoke the vector compute interfaces provided by an external code. GENEXT devices can have up to one thousand external connecting nodes, simplifying multiport coupling implementation [13].

According to [13], Xyce™ calls upon every device (including the GENEXT device) to load the DAE vectors on every Newton step of every nonlinear solve. [13] therefore recommends using the GENEXT coupling method for external problems that are already linear, can be easily linearised over short time scales, or for which a simplified surrogate can be derived that is valid over short time scales. Although our ROM satisfies this condition, two aspects nonetheless make the interface unattractive here.

To our knowledge, the API has no documented restart or checkpointing mechanism. While there is a `simulateUntil` method that could be used to stop the simulation at a given synchronisation point T_n , it is not possible to “rewind” it through the API. We could still obtain a successful implementation through the file-based restarting capability discussed in Section 4.3.1 by re-initialising the driver from a checkpoint. This workaround reintroduces the very I/O the approach was meant to avoid, however.

The second aspect an implementation using the GENEXT interface is unattractive is the interface condition. Therein, the field’s correction of the *previous* iteration $\Delta \mathbf{v}_F^{(k)}$ or $\Delta \mathbf{i}_F^{(k)}$ is replayed. Hence, one would have to

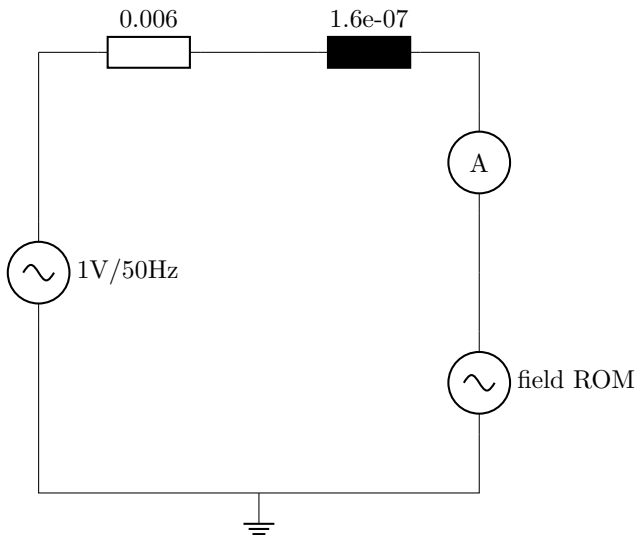


employ the GENEXT device as a lagged source, which forgoes the tight monolithic coupling the interface exists to provide, leaving no advantage over the behavioural-source scheme.

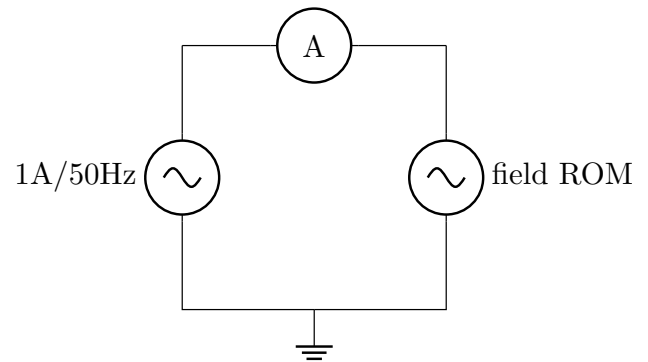
We therefore did not implement the co-simulation using the GENEXT API.

5 Benchmark and Verification

We tested our implementation against many problems. Six of them are stored as presets in the simulation suite and will be discussed in this chapter. We verified the solutions against a monolithic Xyce™ solve, where the “real” parameters of our dummy field subsystem, L_{FEM} and R_{FEM} , replaced the interface ROM plus correction source.



(a) Schematic of the first preset.



(b) Schematic of the second preset.

Figure 5.1: Circuit schematics of the first two presets featuring low frequency sinusoidal sources. The ammeter measures the interface current i_C . They are schematics of the accumulated interface. If the naive interface is used, field ROM is comprised of only the lagged correction term and the ROM is inserted as circuit elements R_{rom} , L_{rom} in the netlist, see Chapter 4.3.2.

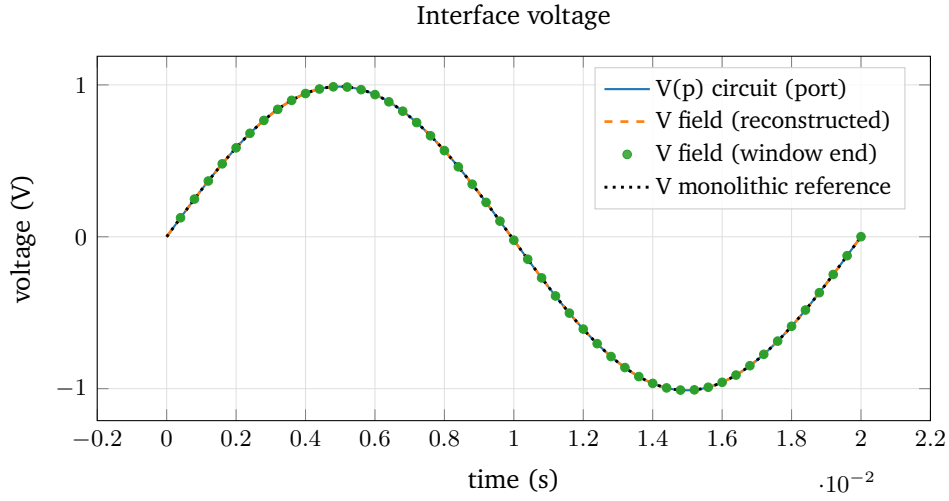
Example 5.1 (LF sources). In this example, we connected a low frequency (LF) source directly at the terminals of the field-circuit interface, see. Figure 5.1. Though they do require different simulation parameters, both interface conditions were able to produce converged solutions that were accurate with respect to the reference (monolithic solution). Figure 5.2 shows a converged solution to the first preset, see Figure 5.1a.

For large enough sample sizes of V_p and/or many $N_{field_eval_intervals}$, the accumulated interface condition will not converge, while the opposite is true for the naive interface condition, which will not converge if the sampling and/or number of field solves is too small. This is a pattern we saw throughout all test cases. We propose an explanation in Chapter 6.

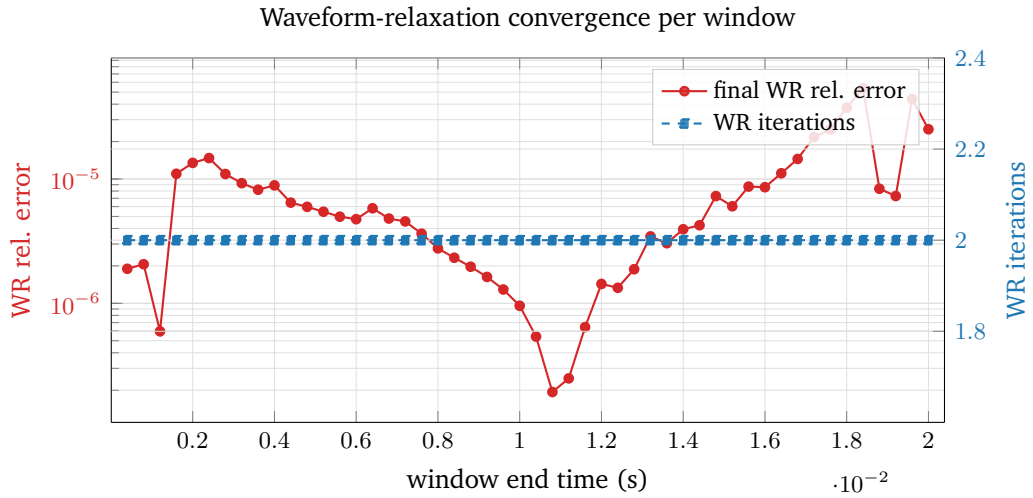
In the edge case of no series circuit components, i.e. $R_s = 0$, $L_s = 0$, see Figure 5.1b, the current-driven coupling showed faster convergence and a smaller error in the convergence criterion than the voltage-driven alternative. This is expected, as imposing the current of an ideal current source on the interface is natural.

Example 5.2 (Higher frequency switching sources). We then tested our implementation against higher frequency problems with switching events. Here, the WR iteration is most likely to fail at the time of the switching event, where the dynamics are strongest. While the backward EULER discretisation of the interface continues to deliver reliable runs on a large set of parameters, the accumulated discretisation converges only on a small set of parameters.

We were able to find suitable parameters, where both discretisations converged and only the convergence metric was different. Figures 5.4a and 5.4b show these converged solutions of the interface voltage. The solution of the accumulated interface discretisation is clearly less accurate than the naive discretisation when compared to the monolithic reference. Though one may speculate, that the amplitude of the oscillation and therefore the solution's accuracy would improve with smaller window sizes (more windows), this was not the case and resulted in divergence in our experiments.



(a) Interface voltage of the converged solution. $V(p)$ is the reconstructed circuit port voltage, containing 201 points per window here. $V(\text{field})$ is the reconstructed field port voltage, containing 11 points per window here.



(b) WR iterations and relative error for each window.

Figure 5.2: Plots of the converged solution to the first preset using the “naive” backward EULER interface discretisation with true field values $L = 1.68 \text{ mH}$, $R = 0.5 \text{ m}\Omega$ and ROM values $L_{\text{ROM}} = 1 \text{ mH}$, $R_{\text{ROM}} = 0.4 \text{ m}\Omega$. The 20 ms time span is divided into 50 windows and Xyce’s solution was resampled to 200 points per window. The voltage-driven field was evaluated 10 times each window.

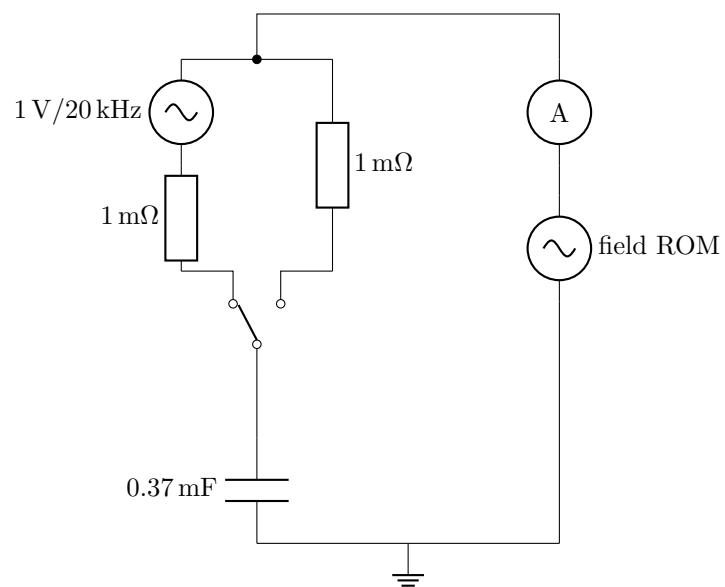
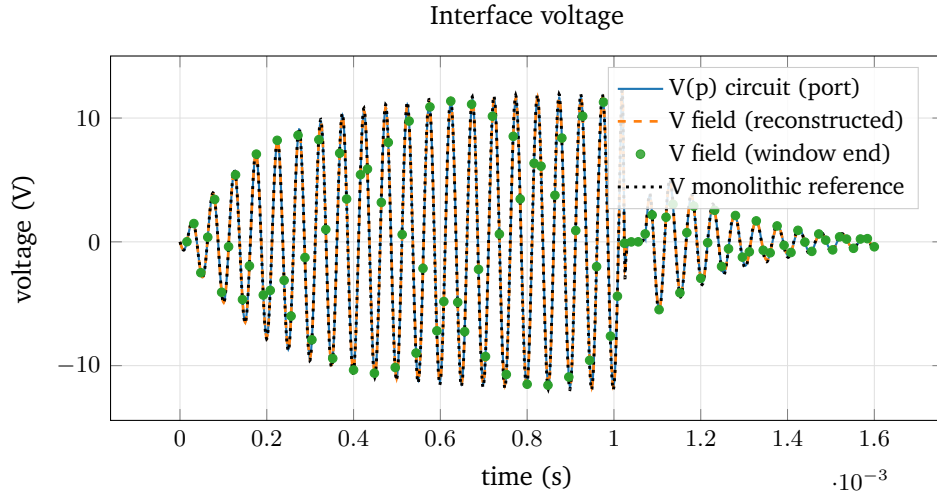
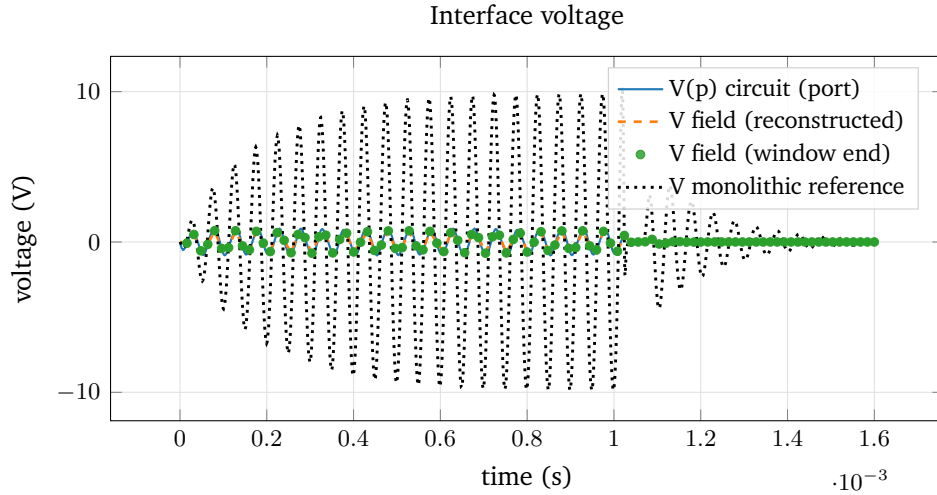


Figure 5.3: Circuit schematic of the third preset. The RLC circuit is being driven by a 1 V/20 kHz source that gets switched off (shorted over 1 mΩ) at 1 ms. Here depicted with the accumulated interface.



(a) Interface discretised with backward EULER. Xyce's solution was resampled to 200 points per window. The voltage-driven field was evaluated 10 times each window



(b) Interface discretised with the accumulated method. Xyce's solution was resampled to one point per window. The voltage-driven field was evaluated once each window.

Figure 5.4: Plots of the converged interface voltage to the third preset using the different interface discretisations with true field values $L = 168 \text{ nH}$, $R = 0.5 \text{ m}\Omega$ and ROM values $L_{\text{ROM}} = 150 \text{ nH}$, $R_{\text{ROM}} = 0.4 \text{ m}\Omega$. The 1.6 ms time span is divided into 100 windows. Voltage-driven field.



6 Application to General Circuits

Parameterstudien und -Empfehlungen.

7 Conclusions and Outlook

Zusammenfassung der Ergebnisse, Schlussfolgerung und Ausblick. Keine neuen Informationen mehr.

7.1 Limitations

The coupling scheme developed in this work makes no assumption about how the field model is formulated or discretised internally, at the cost of not demonstrating the method on an industrial machine model. The latter is left to future work. Considering anything other than two-terminal distributed field devices would require substantial changes to the codebase. The ROM values are constant across the simulation, such that changes of state in the distributed device, e.g. by saturation, are not accurately modelled by the ROM. The implementation only covers transient simulation.

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